

Final Report for CSE 406 Project : TCP reset attack on video streaming

Abrar Rahman Abir (2005072), S. M Kausar Parvej (2005076)

July 2025

Step-by-Step: TCP Reset Attack on YouTube

1. Preparation

- Attacker and victim are connected to the same WiFi/subnet.
- Attacker has root/admin privileges and required tools installed (`scapy`, `arp-scan`, etc.).

2. Discover Network Information

- Find the victim's IP address.
- Find the gateway/router IP address.
- Find the attacker's network interface name.

3. Verify ARP Table

- On the victim, run `arp -a` to see the current MAC for the gateway.

4. Start ARP Spoofing (Poisoning)

- The attacker sends fake ARP replies to the victim and the gateway:
 - Tells the victim: "Gateway is at attacker's MAC."
 - Tells the gateway: "Victim is at attacker's MAC."
- **Result:** All traffic between victim and gateway is routed through the attacker (MITM).

5. Start Packet Monitoring

- The attacker sniffs all packets between the victim and the internet.
- The script looks for YouTube video traffic.

6. Inject TCP RST Packets

- When a YouTube TCP connection is detected:

- The attacker forges TCP packets with the RST (reset) flag, using the correct sequence/ack numbers.
- RST packets are sent to both the victim and the YouTube server, forcibly closing the connection.

7. Observe the Effect

- On the victim, the YouTube video should buffer, stop, or show a network error.
- The attacker's script logs RST injections and YouTube traffic.

8. Stop the Attack and Restore Network

- The attacker stops ARP poisoning.
- The script restores the original ARP tables on the victim and gateway.
- The network returns to normal.

A Successful Attack

Yes, the attack is successful. Here's why:

1. ARP Table Manipulation (`arp -a`):

- On the victim machine, running `arp -a` during the attack shows that the MAC address for the gateway has changed to the attacker's MAC address.
- This demonstrates that ARP poisoning is effective, and the attacker is acting as a man-in-the-middle, intercepting and injecting packets between the victim and the gateway.

2. TCP Reset Evidence (`netstat -s`):

- Running `netstat -s` on the victim before and after the attack shows an increase in the TCP reset counter.
- This confirms that the victim is receiving TCP RST (reset) packets, which are being injected by the attacker.

3. User-Visible Disruption (YouTube Video Stops):

- While the attack is running, the YouTube video on the victim buffers, stops, or shows a network error.
- This is a direct, observable effect of the TCP reset attack forcibly closing the video streaming connections.

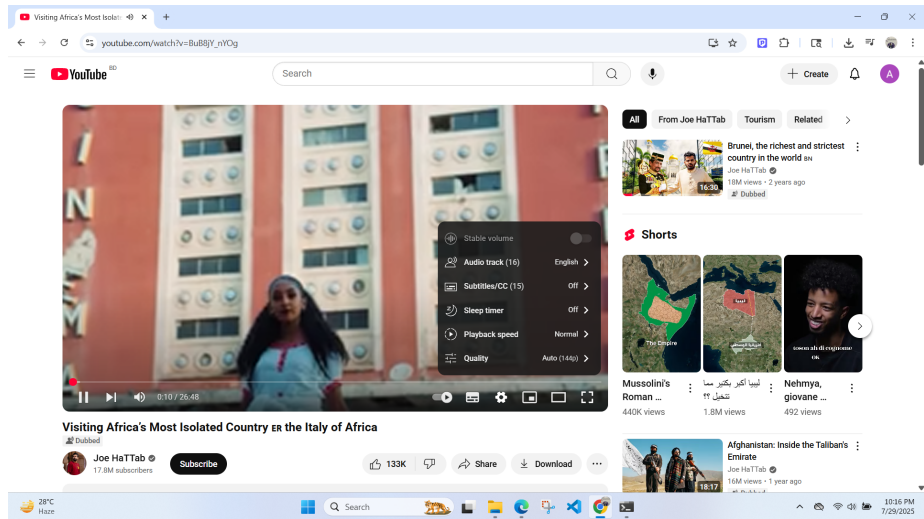


Figure 1: Normal YouTube Behavior

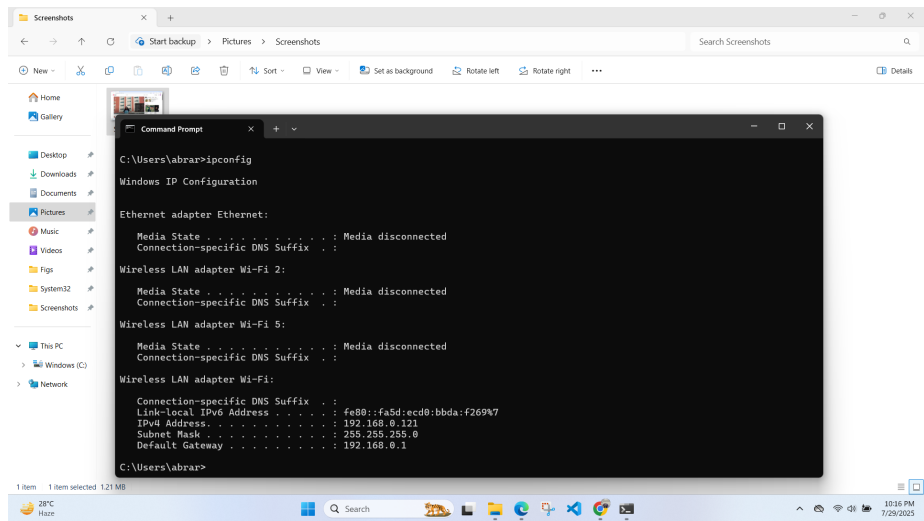


Figure 2: Victim PC's IP

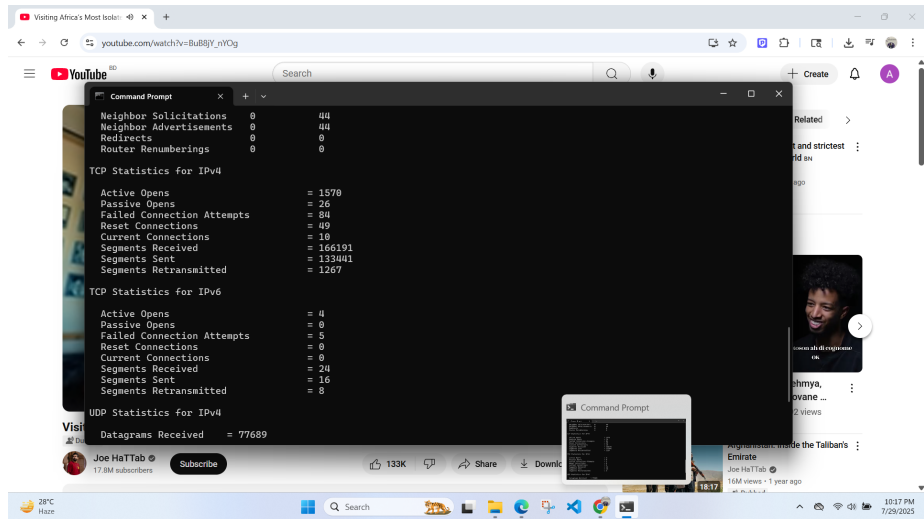


Figure 3: Initial TCP reset connections.

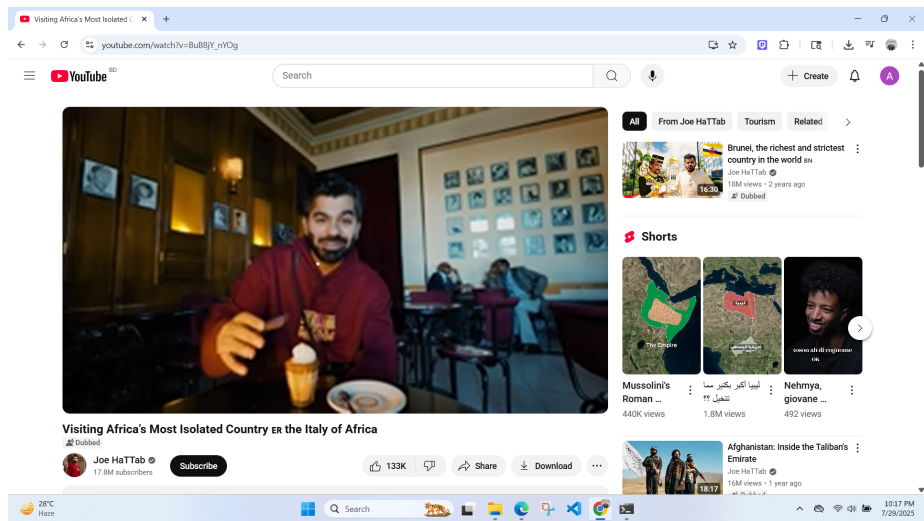


Figure 4: Under Attack, video does not play.

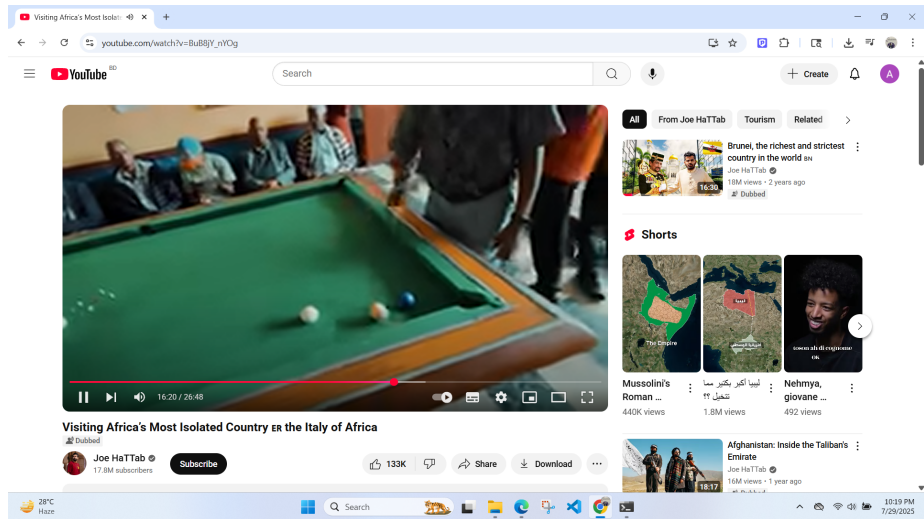


Figure 5: Attack finished. Video plays.

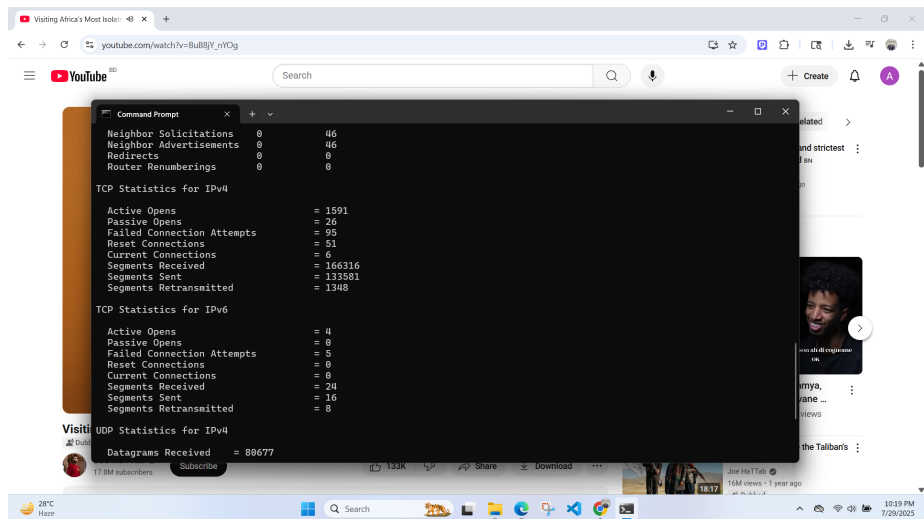


Figure 6: Increased TCP reset connections.



Task	Description	Student ID
Detection, Automation, and Demo Scripting	Developed scripts for automatic victim/gateway discovery, aggressive attack automation, and user-friendly demo execution. Enhanced visible disruption for demo.	72 + 76
Testing, Validation, and Analysis	Conducted extensive testing on devices and setups. Validated attack using tools (<code>arp -a</code> , <code>netstat -s</code>), observed video disruption, and analyzed results.	72 + 76
Documentation and Reporting	Prepared detailed documentation, guides, and academic reporting. Included step-by-step instructions, troubleshooting, and evidence of attack success.	72 + 76

Table 1: Team Member Contributions and Student IDs